# Comprehensive Examination

Directions: Each of the numbered items or incomplete statements in this section is followed by answers or by completions of the statement. Select the **one**-lettered answer or completion that is **best** in each case.

**1.** A young man is brought to the emergency department after being mugged. He was stabbed in the shoulder after refusing to give his wallet to his assailant. If the stab wound lacerates the posterior humeral circumflex artery passing through the quadrangular space on the shoulder region, which of the following nerves might be injured?

**(A)** Radial nerve
**(B)** Axillary nerve
**(C)** Thoracodorsal nerve
**(D)** Suprascapular nerve
**(E)** Accessory nerve

**2.** A victim of an automobile accident is unable to abduct her left arm. This indicates damage to which of the following parts of the brachial plexus?

**(A)** Middle trunk and posterior cord
**(B)** Middle trunk and lateral cord
**(C)** Lower trunk and lateral cord
**(D)** Upper trunk and posterior cord
**(E)** Lower trunk and medial cord

**3.** A biomedical engineer would like to reconstruct the arm of a boy who underwent amputation to treat a life-threatening infection. In designing the prosthetic arm, the engineer will need to know that which of the following muscles flexes the elbow and is innervated by the radial nerve?

**(A)** Flexor digitorum longus
**(B)** Brachioradialis
**(C)** Brachialis
**(D)** Extensor digitorum longus
**(E)** Biceps brachii

**4.** Young Johnny was playing on the playground at school when he fell and struck his arm against the swing set. He ran to the school nurse, complaining of which of the following conditions as a result of injuring the radial nerve in the spiral groove of the humerus?

**(A)** Numbness over the medial side of the forearm
**(B)** Inability to oppose the thumb
**(C)** Weakness in pronating the forearm
**(D)** Weakness in abducting the arm
**(E)** Inability to extend the hand

**5.** An indoor soccer player runs into another player while running after the ball. She falls to the ground and fractures the medial epicondyle of the humerus. Which of the following symptoms might she present with when seeing the physician in the emergency department?

**(A)** Impaired abduction of the hand
**(B)** Carpal tunnel syndrome
**(C)** Wrist drop
**(D)** Thenar atrophy
**(E)** Inability to sweat on the medial part of the hand

**6.** After winning a boxing match, a 24-year-old man is unable to abduct his fingers. Which of the following nerves is injured?

**(A)** Ulnar nerve
**(B)** Median nerve
**(C)** Radial nerve
**(D)** Musculocutaneous nerve
**(E)** Axillary nerve

**7.** A 42-year-old woman presents to an outpatient clinic with a 6-month history of numbness and tingling on the palmar aspect of her lateral three and one-half fingers, loss of pronation, and flattening of the thenar eminence. Injury to which of the following nerves could cause such a condition?

**(A)** Axillary nerve
**(B)** Musculocutaneous nerve
**(C)** Median nerve
**(D)** Radial nerve
**(E)** Ulnar nerve

**8.** A ballet dancer falls to the floor and hurts herself during a practice session before opening night. She sustains an injury to the thoracodorsal nerve that would probably affect the strength of which of the following movements?

**(A)** Adduction of the scapula
**(B)** Elevation of the scapula
**(C)** Abduction of the arm
**(D)** Extension of the arm
**(E)** Lateral rotation of the arm

**9.** A 23-year-old man falls from a ladder and injures his arm. On examination, he feels numbness and has no sweating on the lateral side of his forearm, suggesting damage to the lateral antebrachial cutaneous nerve. The cell bodies of injured nerve fibers involved in sweating are located in which of the following structures?

**(A)** Collateral ganglia
**(B)** Dorsal root ganglia
**(C)** Sympathetic chain ganglia
**(D)** Lateral horn of spinal cord
**(E)** Anterior horn of spinal cord

**10.** There are only 30 minutes left before the concert starts. The pianist accidentally cuts herself over the palmar surface of her wrist and notices that she is unable to pick up a piece of music between her index and middle fingers. Which of the following nerves is most likely damaged?

**(A)** Radial nerve
**(B)** Axillary nerve
**(C)** Ulnar nerve
**(D)** Median nerve
**(E)** Anterior interosseous nerve

**11.** A 29-year-old carpenter receives a crush injury to his metacarpophalangeal joint of the fourth digit (ring finger) while remodeling his neighbor's porch. Which of the following pairs of nerves innervates the muscle that moves the injured joint?

**(A)** Median and ulnar nerves
**(B)** Radial and median nerves
**(C)** Musculocutaneous and ulnar nerves
**(D)** Ulnar and radial nerves
**(E)** Radial and axillary nerves

**12.** A 21-year-old man celebrating his birthday gets a little carried away with his friends and starts a bar fight. He is stabbed with a knife that severs the roots of C5 and C6 of the brachial plexus. Which of the following muscles is likely to be paralyzed?

**(A)** Infraspinatus
**(B)** Flexor carpi ulnaris
**(C)** Palmar interossei
**(D)** Adductor pollicis
**(E)** Palmaris brevis

**13.** The secretary of a rather verbose academic physician in internal medicine complains of numbness and tingling in her hands and fingers. She is constantly typing long patient visit dictations and now has carpal tunnel syndrome, which is due to compression of which of the following structures?

**(A)** Ulnar artery
**(B)** Ulnar nerve
**(C)** Median nerve
**(D)** Flexor carpi radialis tendon
**(E)** Palmaris longus tendon

**14.** While playing in a Super Bowl game, a 26-year-old professional football player is tackled, and his anterior cruciate ligament is torn. Although not injured, the anterior cruciate ligament of the knee joint:

**(A)** Becomes taut during flexion of the leg
**(B)** Resists posterior displacement of the femur on the tibia
**(C)** Inserts into the medial femoral condyle
**(D)** Helps prevent hyperflexion of the knee joint
**(E)** Is lax when the knee is extended

**15.** A man interviewing for a new administrative position as hospital chief executive officer notices difficulty walking after sitting with his leg crossed for 2 hours. He was nervous during the interview but even more so now that he is attempting to stand to follow two board members for a tour of the hospital. Which of the following actions is most seriously affected by compression and temporary paralysis of the deep peroneal nerve?

**(A)** Plantar flexion of the foot
**(B)** Dorsiflexion of the foot
**(C)** Abduction of the toes
**(D)** Adduction of the toes
**(E)** Inversion of the foot

**16.** Deep venous thrombosis is a common complication from sitting in one position for

a prolonged duration, such as during a long car trip or a long plane flight. The first vascular channels likely to be obstructed or occluded by an embolus from the deep veins of a lower limb are the:

**(A)** Tributaries of the renal veins
**(B)** Branches of the coronary arteries
**(C)** Sinusoids of the liver
**(D)** Tributaries of the pulmonary veins
**(E)** Branches of the pulmonary arteries

**17.** During recruitment by the local representative for the Marines, a young college student presents with the condition known as flat foot. His foot is displaced laterally and everted, and the head of the talus is no longer supported. Which of the following ligaments probably is stretched?

**(A)** Plantar calcaneonavicular (spring)
**(B)** Calcaneofibular
**(C)** Anterior talofibular
**(D)** Plantar calcaneocuboid (short plantar)
**(E)** Anterior tibiotalar

**18.** During a sports medicine physical by a local family physician, a young woman is tested for stability of her joints before tryouts for the high school team. Which of the following ligaments is important in preventing forward displacement of the femur on the tibia when the weight-bearing knee is flexed?

**(A)** Medial meniscus
**(B)** Tibial collateral ligament
**(C)** Fibular collateral ligament
**(D)** Posterior cruciate ligament
**(E)** Anterior cruciate ligament

**19.** A 21-year-old man falls from the attic and is brought to the emergency department. Examination and x-rays reveal that the lateral longitudinal arch of his foot is flattened. Which of the following bones is displaced?

**(A)** Talus
**(B)** Medial three metatarsals
**(C)** Navicular
**(D)** Cuneiform
**(E)** Cuboid

**20.** A 72-year-old woman with Parkinson disease fell down in the bathtub at her home and suffered a dislocation of the hip joint that may result in vascular necrosis of the femoral head and neck because of injuries to the arteries. Which of the following arteries might remain intact?

**(A)** Lateral femoral circumflex artery
**(B)** Medial femoral circumflex artery
**(C)** Obturator artery
**(D)** Inferior gluteal artery
**(E)** Middle sacral artery

**21.** A 78-year-old woman undergoes knee surgery because her lateral meniscus is torn. Before injury, the normal lateral meniscus of the knee joint:

**(A)** Is C-shaped or forms a semicircle
**(B)** Is attached to the fibular collateral ligament
**(C)** Is larger than the medial meniscus
**(D)** Lies outside the synovial cavity
**(E)** Is more frequently torn in injuries than the medial meniscus

**22.** A 17-year-old boy is involved in a group fight, and a stab wound lacerates a ventral root of his thoracic spinal nerve. Cell bodies of the injured nerve fibers are located in which of the following nervous structures?

**(A)** Dorsal root ganglia and sympathetic trunk
**(B)** Lateral horn of spinal cord and dorsal root ganglia
**(C)** Anterior horn and lateral horn of spinal cord
**(D)** Sympathetic trunk and lateral horn of spinal cord
**(E)** Anterior horn of spinal cord and sympathetic trunk

**23.** A race car driver is brought to the city trauma center after a high-speed crash in which his car spun out of control and struck a concrete embankment. He has blunt trauma to his chest and undergoes extensive vascular studies to determine which blood vessels are still intact. The interventional radiologist recalls that one of the following veins drains directly into the superior vena cava. Which vein would this be?

**(A)** Internal thoracic vein
**(B)** Azygos vein
**(C)** Hemiazygos vein
**(D)** Right superior intercostal vein
**(E)** Left superior intercostal vein

**24.** A 58-year-old stockbroker is brought to the cardiac catheterization laboratory emergently after evaluation in the emergency department

has determined that he is suffering from an acute myocardial infarction. During the catheterization, he is found to have inadequate blood flow in the artery that runs along the great cardiac vein in the anterior interventricular sulcus of the heart. This is most likely an acute occlusion of the:

**(A)** Circumflex branch of the left coronary artery
**(B)** Marginal branch of the right coronary artery
**(C)** Left coronary artery
**(D)** Right coronary artery
**(E)** Posterior interventricular artery

**25.** A retired teacher suffers from a massive heart attack while playing golf and dies in the intensive care unit. Autopsy reveals the cause of death as severely diminished blood flow in the coronary arteries. This most likely resulted from embolization of an atherosclerotic plaque at the origin of which of the following vascular structures?

**(A)** Pulmonary trunk
**(B)** Ascending aorta
**(C)** Coronary sinus
**(D)** Descending aorta
**(E)** Aortic arch

**26.** A 21-year-old woman comes to the emergency department with acute chest pain and shortness of breath. Her chest x-ray shows opacification of one of her lungs. She undergoes thoracentesis, which reveals that she has a chylothorax resulting from rupture of the thoracic duct. Lymphatic drainage remains normal in which of the following areas?

**(A)** Left thorax
**(B)** Right thorax
**(C)** Left abdomen
**(D)** Right pelvis
**(E)** Left lower limb

**27.** An elderly man is choking on his food at a restaurant, and attempts by other patrons to dislodge the food bolus using the Heimlich procedure have failed. A retired anesthesiologist rushes to his table and prepares for emergent tracheotomy. She locates the manubrium of the sternum and recalls that it is free from articulation with which of the following structures?

**(A)** Body of the sternum
**(B)** First rib
**(C)** Second rib
**(D)** Third rib
**(E)** Clavicle

**28.** A stab wound penetrates the posterior thoracic wall near vertebra of a 24-year-old man. Examination at the emergency department indicates a lesion of gray rami communicantes. Which of the following nerve fibers would most likely be damaged?

**(A)** General somatic afferent (GSA) fibers
**(B)** Postganglionic parasympathetic fibers
**(C)** Preganglionic sympathetic fibers
**(D)** Postganglionic sympathetic fibers
**(E)** General visceral afferent (GVA) fibers

**29.** A 31-year-old NHL hockey player complains of numbness in the area of his umbilicus after the national championship game. Which of the following structures that carries GSA fibers was injured during the hockey game?

**(A)** Sympathetic trunk
**(B)** Dorsal root
**(C)** Greater splanchnic nerve
**(D)** Gray rami communicantes
**(E)** White rami communicantes

**30.** A 75-year-old veteran suffers a heart attack and is found in his home unconscious. He is in ventricular tachycardia and is shocked into normal sinus rhythm. He undergoes emergent catheterization and is found to have thrombosis in the coronary sinus. Which of the following cardiac veins might remain normal in diameter by catheterization study by the cardiologist?

**(A)** Great cardiac vein
**(B)** Middle cardiac vein
**(C)** Anterior cardiac vein
**(D)** Small cardiac vein
**(E)** Oblique cardiac vein

**31.** A 35-year-old man is suffering from an infected mediastinum (mediastinitis) after neck and chest injuries resulting from a head-on automobile collision. He has been intubated since the accident and on broad-spectrum intravenous antibiotics since admission; however, the infection continues to progress throughout the mediastinum. Which of the following structures is free from infection?

**(A)** Thymus gland
**(B)** Esophagus
**(C)** Trachea
**(D)** Lungs
**(E)** Heart

**32.** A 42-year-old man who has a rare neurologic disorder comes to the emergency department. On examination, he is unable to protrude his tongue. Which of the following muscles is paralyzed?

**(A)** Hyoglossus
**(B)** Genioglossus
**(C)** Styloglossus
**(D)** Palatoglossus
**(E)** Geniohyoid

**33.** A new biotech company is interested in developing a new mechanical heart with a superficial implantable and rechargeable battery for easy access. During the design phase, the physician hired from the local academic hospital is asked about which structure carries or comes in contact with oxygenated blood. Which of the following answers is correct?

**(A)** Pectinate muscle
**(B)** Crista terminalis
**(C)** Septomarginal trabecula
**(D)** Pulmonary vein
**(E)** Pulmonary artery

**34.** A 62-year-old man is diagnosed with a Pancoast tumor that invades the inferior trunk of the brachial plexus. Which of the following muscle actions most likely resulted from injury to the brachial plexus?

**(A)** Lateral rotation of the arm
**(B)** Extension of the ring finger
**(C)** Abduction of the index finger
**(D)** Flexion of the forearm
**(E)** Pronation of the forearm

**35.** A 67-year-old woman complains of increasing urinary frequency and a heaviness in her pelvic area. On examination, her uterine cervix is visible at the vaginal opening. Which of the following conditions causes this symptom?

**(A)** Tear of the transversalis fascia
**(B)** Weakness of the ovarian ligament
**(C)** Relaxation of the cardinal ligament
**(D)** Weakness of the arcuate pubic ligament
**(E)** Paralysis of the piriformis muscle

**36.** Weight lifters in competition are often concerned about muscle tone and complications with hernias. In particular, the most common hernia in this case is an indirect inguinal hernia, which appears:

**(A)** Lateral to the inferior epigastric artery
**(B)** Between the inferior epigastric and obliterated umbilical arteries
**(C)** Medial to the obliterated umbilical artery
**(D)** Between the median and medial umbilical folds
**(E)** Between the linea alba and linea semilunaris

**37.** A 32-year-old man is involved in a car accident and suffers from a crushed internal injury in his abdomen. Examination reveals a lesion of parasympathetic fibers in the vagus nerve, which interferes with glandular secretory or smooth muscle functions in which of the following organs?

**(A)** Bladder
**(B)** Transverse colon
**(C)** Sigmoid colon
**(D)** Prostate gland
**(E)** Rectum

**38.** Pancreatic cancer has one of the highest mortality rates of all cancers because of the lack of symptoms until an advanced stage of disease. The one exception is the cancer that is slow growing and located in the head of the pancreas. This may present in an early stage by causing compression of which of the following structures?

**(A)** Duodenojejunal junction
**(B)** Gastroduodenal artery
**(C)** Bile duct
**(D)** Inferior mesenteric artery
**(E)** Common hepatic duct

**39.** An elderly man with a known large abdominal aortic aneurysm presents to the emergency department with acute severe and diffuse pain in his abdomen. The physician performing the evaluation considers mesenteric ischemia, a life-threatening disease, as a possible etiology. Which of the following organs may be spared from ischemia in the presence of an occlusive lesion in the celiac trunk?

**(A)** Liver
**(B)** Spleen
**(C)** Pancreas
**(D)** Gallbladder
**(E)** Stomach

**40.** A young woman with cryptogenic cirrhosis presents to the university hospital for an evaluation as a possible candidate for liver transplant. She has late-stage cirrhosis, and her liver–spleen scan shows a high degree of portal hypertension. The portal venous system includes which of the following veins?

**(A)** Left suprarenal vein
**(B)** Inferior epigastric vein
**(C)** Superior rectal vein
**(D)** Azygos vein
**(E)** Hepatic vein

**41.** A 2-month-old boy is admitted to the hospital with diarrhea, vomiting, and difficulty in bowel movements. A computed tomography (CT) scan of his abdomen reveals a distinctly dilated lower colon and is diagnosed as having Hirschsprung disease. Which of the following is the most likely embryonic mechanism responsible for development of this congenital megacolon?

**(A)** Malrotation of the hindgut
**(B)** Abnormal formation of the urorectal septum
**(C)** Failure of neural crest cell migration into the hindgut
**(D)** Absence of visceral afferent cells in the myenteric plexus
**(E)** Failure of partitioning of the cloaca

**42.** A 29-year-old farmer falls on tractor blades and injures his groin. Several days later, he comes to the emergency department and examination by the physician reveals that the urogenital diaphragm and bulbourethral glands are infected. The infected deep perineal space:

**(A)** Is formed superiorly by the perineal membrane
**(B)** Is formed inferiorly by Colles' fascia
**(C)** Contains a segment of the dorsal nerve of the penis
**(D)** Contains superficial transverse perineal muscles
**(E)** Contains the greater vestibular glands

**43.** A 2-day-old baby is admitted to the hospital with vomiting, fever, and rectal bleeding. Physical and CT scan examinations reveal small bowel obstruction and are diagnosed as having Meckel diverticulum. Which portion of the gastrointestinal tract is most likely involved in the development of the diverticulum that may be connected to the inner surface of the umbilicus by a fibrous cord?

**(A)** Stomach
**(B)** Jejunum
**(C)** Ileum
**(D)** Appendix
**(E)** Sigmoid colon

**44.** A 59-year-old woman has a large pelvic tumor that compresses the inferior hypogastric (pelvic) plexus. Parasympathetic nerve fibers in this plexus come from which of the following nerves?

**(A)** Lumbar splanchnic nerves
**(B)** Pelvic splanchnic nerves
**(C)** Sacral sympathetic chain ganglia
**(D)** Vagus nerve
**(E)** Sacral splanchnic nerves

**45.** Because of a lesion, the parasympathetic nerve fibers are unable to induce a contraction of the detrusor muscle and relaxation of the internal sphincter. The injured parasympathetic fibers that supply the urinary bladder are derived from which of following nerves?

**(A)** Vagus nerve
**(B)** Pelvic splanchnic nerve
**(C)** Sacral splanchnic nerve
**(D)** Lesser splanchnic nerve
**(E)** Greater splanchnic nerve

**46.** At a local hospital tumor board, a gynecologic oncologist discusses the next case for the multidisciplinary team. He explains the rationale for using chemotherapy and radiation after surgical resection because carcinoma of the uterus can spread directly to the labium majus through lymphatics that follow the:

**(A)** Ovarian ligament
**(B)** Suspensory ligament of the ovary
**(C)** Round ligament of the uterus
**(D)** Uterosacral ligaments
**(E)** Pubocervical ligaments

**47.** A young couple is seeing a sex therapist for the first time to determine the cause of some of their recent difficulties. The husband tells the therapist that he no longer has sensation in his scrotum after a race car accident. Which of the following nerves carries undamaged sensory nerve fibers?

**(A)** Ilioinguinal nerve
**(B)** Genitofemoral nerve
**(C)** Iliohypogastric nerve
**(D)** Perineal branch of the pudendal nerve
**(E)** Perineal branch of the posterior femoral cutaneous nerve

**48.** A patient has a damaged pelvic outlet as the result of an automobile accident. Following this accident, which of the following structures is still intact?

**(A)** Sacrotuberous ligament
**(B)** Inferior pubic ramus
**(C)** Pubic crest
**(D)** Ischial tuberosity
**(E)** Coccyx

**49.** A forensic pathologist is examining the pelvic bone of a murder victim to identify the sex of the victim. Which of the following characteristics is that of a female pelvis?

**(A)** Oval-shaped pelvic inlet
**(B)** Smaller pelvic outlet
**(C)** Lesser pubic angle
**(D)** Narrower and longer sacrum
**(E)** Narrower and deeper pelvic cavity

**50.** A 26-year-old woman experiences severe back pain from an automobile accident. A CT scan reveals that the L5 vertebral foramen is completely obliterated by a collapsed L5 laminae and pedicles. In this injury, which of the following structures is crushed?

**(A)** Vertebral artery
**(B)** Spinal cord
**(C)** Filum terminale externus (filum of the dura)
**(D)** Denticulate ligament
**(E)** Cauda equina

**51.** A performer at the traveling circus for the state fair has injured her shoulder during a routine. When she fell off the trapeze, she struck the ground on her back, and most of the blunt force was directed toward her shoulders. A crush injury of the suboccipital nerve would result in paralysis of which of the following muscles?

**(A)** Splenius capitis
**(B)** Trapezius
**(C)** Rectus capitis posterior major
**(D)** Levator scapulae
**(E)** Iliocostalis

**52.** A neonate is brought to the pediatrician with headaches, fevers, and change in mental status, mostly manifesting as lethargy and lack of appetite. The baby is admitted emergently to rule out meningitis and is to undergo a diagnostic lumbar puncture. When withdrawing cerebrospinal fluid (CSF), the needle may penetrate which of the following pairs of structures?

**(A)** Dura mater and denticulate ligament
**(B)** Arachnoid mater and pia mater
**(C)** Dura mater and arachnoid mater
**(D)** Annulus fibrosus and pia mater
**(E)** Arachnoid mater and nucleus pulposus

**53.** Mrs. Jones was riding in the front seat of her son's van when the vehicle abruptly stopped. Unfortunately, she was not wearing her seatbelt and was thrown forward. As a result of the accident, the transverse processes of her cervical vertebrae were crushed against the dashboard of the van. Which of the following muscles might be paralyzed?

**(A)** Trapezius
**(B)** Latissimus dorsi
**(C)** Rhomboid major
**(D)** Levator scapulae
**(E)** Serratus posterior superior

**54.** A middle-aged woman is receiving a shiatsu massage when her therapist notices that the client has numbness in her neck. A lack of sensation of the skin over the anterior triangle of the neck may be due to injury to which of the following nerves?

**(A)** Great auricular nerve
**(B)** Transverse cervical nerve
**(C)** Superior ramus of the ansa cervicalis
**(D)** Inferior ramus of the ansa cervicalis
**(E)** Superior laryngeal nerve

**55.** A young man presents to your clinic with a 3-month history of an enlarging mass in his right neck. He undergoes a CT scan that reveals a fluid-filled mass where the common carotid artery bifurcates. Which of the following is another structure that would be located at this level?

**(A)** Thyroid isthmus
**(B)** Cricoid cartilage
**(C)** Sternal angle
**(D)** Superior border of the thyroid cartilage
**(E)** Jugular notch

**56.** An ultimate fighter is brought into the locker room after being knocked out by his opponent. He has suffered severe trauma to the articular disk and capsule of the temporomandibular joint. This could result in paralysis of which of the following muscles?

**(A)** Masseter
**(B)** Temporalis
**(C)** Medial pterygoid
**(D)** Lateral pterygoid
**(E)** Buccinator

**57.** A 25-year-old woman is suffering from facial paralysis and exhibits ptosis (drooping of the upper eyelid). Injury to which of the following nerves would result in ptosis?

**(A)** Trochlear nerve
**(B)** Abducens nerve
**(C)** Oculomotor nerve
**(D)** Ophthalmic nerve
**(E)** Facial nerve

**58.** A 67-year-old man is known to have an infection in the superior petrosal sinus. The infected sinus lies in the margin of which of the following structures?

**(A)** Tentorium cerebelli
**(B)** Falx cerebri
**(C)** Falx cerebelli
**(D)** Diaphragma sellae
**(E)** Straight sinus

**59.** A 59-year-old man has pus in the loose connective tissue layer of the scalp, and consequently, his superior sagittal sinus is infected. The arachnoid granulations in the infected sinus:

**(A)** Absorb CSF into the dural venous sinuses
**(B)** Are storage areas for CSF
**(C)** Produce CSF
**(D)** Allow CSF to return to the ventricles of the brain
**(E)** Filter venous blood into CSF

**60.** While speaking at a charity program, an elderly woman suddenly collapses to the ground. She is diagnosed as having a massive stroke, but it is unclear where the vascular lesion is located. She undergoes a cerebral arteriogram. Normally, the great cerebral vein of Galen drains directly into which of the following sinuses?

**(A)** Superior sagittal sinus
**(B)** Inferior sagittal sinus
**(C)** Cavernous sinus
**(D)** Transverse sinus
**(E)** Straight sinus

**61.** While resecting a tumor in the palate, the surgical intern removes the mass but finds that it is deeply invasive into the tendon that loops around the pterygoid hamulus. Which of the following muscles would most likely be paralyzed?

**(A)** Tensor tympani
**(B)** Tensor veli palatini
**(C)** Levator veli palatini
**(D)** Superior pharyngeal constrictor
**(E)** Stylohyoid

**62.** During a domestic incident, a 28-year-old woman receives a vertical stab wound that lacerates the pterygomandibular raphe. As a result, which of the following muscles would be paralyzed?

**(A)** Superior and middle pharyngeal constrictors
**(B)** Middle and inferior pharyngeal constrictors
**(C)** Superior pharyngeal constrictor and buccinator muscles
**(D)** Medial and lateral pterygoid muscles
**(E)** Tensor veli palatini and levator veli palatini

**63.** A 31-year-old football player with a head injury is brought to a local emergency department. Physical examination, radiogram, and a magnetic resonance imaging scan indicate a lesion of the trigeminal nerve. Which of the following muscles is most likely paralyzed?

**(A)** Geniohyoid
**(B)** Palatoglossus
**(C)** Cricothyroid
**(D)** Tensor veli palatini
**(E)** Levator veli palatini

**64.** The muscle that indents the submandibular gland and divides it into superficial and deep parts is paralyzed because of compression by a large salivary glandular tumor. Which of the following muscles is involved?

**(A)** Hyoglossus
**(B)** Digastric posterior belly
**(C)** Styloglossus
**(D)** Stylohyoid
**(E)** Mylohyoid

**65.** A young homeless child presents to the emergency department with high fever and low blood pressure. He had strep throat 2 weeks ago when seen in the free clinic; however, his parents did not give him his antibiotic medication. He now has an abscess in the auditory tube that is blocking communication between the nasopharynx and which of the following structures?

**(A)** Vestibule of the inner ear
**(B)** Middle ear
**(C)** Semicircular canals
**(D)** External ear
**(E)** Inner ear

**66.** A man with a neuromuscular autoimmune disease has another attack at home. His wife calls the doctor's office because he is unable to speak. Once at the office, he is unable to open his jaw because of paralysis of which of the following muscles?

**(A)** Medial pterygoid
**(B)** Masseter
**(C)** Temporalis
**(D)** Lateral pterygoid
**(E)** Buccinator

**67.** A 29-year-old baseball player whose head is hit by a ball is brought to an emergency department. Physical examination and radiogram reveal fracture of the temporal bone and damage to the lesser petrosal nerve. Which of the following conditions could occur as a result of injury to the lesser petrosal nerve?

**(A)** Lack of lacrimal secretion
**(B)** Lack of submandibular gland secretion
**(C)** Lack of parotid gland secretion
**(D)** Constriction of the pupil
**(E)** Ptosis of the upper eyelid

**68.** A 26-year-old woman with the flu comes to an outpatient clinic. On examination, a physician diagnoses her with sinus and palate infection, and her uvula deviates to the left side on phonation. Which of the following nerves is injured?

**(A)** Left hypoglossal nerve
**(B)** Right hypoglossal nerve
**(C)** Left vagus nerve
**(D)** Right vagus nerve
**(E)** Left trigeminal nerve

**69.** During a palatine tonsillectomy, a surgeon must ligate arteries to avoid bleeding within the surgical field. Which of the following arteries can be spared?

**(A)** Lesser palatine artery
**(B)** Facial artery
**(C)** Lingual artery
**(D)** Superior thyroid artery
**(E)** Ascending pharyngeal artery

**70.** A young child is evaluated by his pediatrician for chronic nosebleeds. He is referred to an allergist when it is determined that his nasal cavity is chronically dry because of a lack of glandular secretions. A possible etiology may be a lesion of which of the following structures?

**(A)** Superior cervical ganglion
**(B)** Lesser petrosal nerve
**(C)** Facial nerve in the facial canal
**(D)** Greater petrosal nerve
**(E)** Deep petrosal nerve

**Questions 71 and 72 are based on the following case: A 12-year-old girl suffers from a type of neural tube defect called tethered cord syndrome, a congenital anomaly that results from defective closure of the neural tube. This syndrome is characterized by an abnormally low conus medullaris, which is tethered by a short, thickened filum terminale, leading to progressive neurologic defects in the legs and feet.**

**71.** Which of the following defects is commonly associated with the tethered cord syndrome?

**(A)** Spina bifida occulta
**(B)** Kyphosis
**(C)** Meningomyelocele
**(D)** Herniated disk
**(E)** Scoliosis

**72.** This girl has strong muscle function of the flexors of the thigh, but she has weakness of the extensors (hamstrings). A lesion has occurred at which of the following spinal cord levels?

**(A)** T12
**(B)** L1
**(C)** L3
**(D)** L5
**(E)** S5

**73.** A 36-year-old plumber slips and breaks a porcelain sink, cutting an anterior aspect of his wrist deeply on a sharp edge. On arrival to the emergency department, he can adduct his thumb but not oppose it. Which of the following structures has been severed?

**(A)** Radial nerve
**(B)** Median nerve
**(C)** Ulnar nerve
**(D)** Anterior interosseous nerve
**(E)** Posterior interosseous nerve

**74.** A 21-year-old soccer player is tackled via a high-impact sweeping injury about the right knee, causing a posterior knee dislocation. Soon after, he is unable to plantar flex his right ankle or flex his toes. He also experiences loss of sensation on the sole of his right foot. Which of the following nerves is most likely injured?

**(A)** Saphenous nerve
**(B)** Tibial nerve
**(C)** Deep peroneal nerve
**(D)** Superficial peroneal nerve
**(E)** Common peroneal nerve

**75.** A 56-year-old man with a one pack/day history of cigarette smoking is found to have a malignant squamous cell carcinoma located in the superior sulcus of Pancoast (superior apex of the lung). Shortly after the diagnosis is known, the patient develops symptoms that consist of shoulder pain along with ptosis, miosis, enophthalmos, and anhidrosis. Which of

the following nerves is most likely compressed by the tumor mass?

**(A)** Phrenic nerve
**(B)** Vagus nerve
**(C)** Cervical sympathetic trunk
**(D)** Ansa cervicalis
**(E)** Recurrent laryngeal nerve

**76.** A 57-year-old woman presents with increasing numbness of the fourth and fifth digits on her right hand. On examination, it is noticed that she has a wasted hypothenar eminence, inability to abduct the thumb, and a characteristic claw hand. Which of the following nerves has most likely been injured?

**(A)** Axillary nerve
**(B)** Anterior interosseous nerve
**(C)** Radial nerve
**(D)** Ulnar nerve
**(E)** Median nerve

**77.** A 47-year-old woman has had a lumpectomy and axillary dissection to check for metastasis. She has come in for her follow-up appointment, and her physician notices on her physical examination that the medial part of her scapula protrudes from her back and that she is not able to raise her arm above the horizontal level. Which of the following nerves has been damaged during her surgery?

**(A)** Median nerve
**(B)** Ulnar nerve
**(C)** Thoracodorsal nerve
**(D)** Long thoracic nerve
**(E)** Radial nerve

**78.** A 31-year-old carpenter cuts his left arm with a utility knife and is brought to a local emergency department. He complains of numbness on the medial side of his arm. Which of the following nerves is most likely injured?

**(A)** Axillary nerve
**(B)** Musculocutaneous nerve
**(C)** Medial brachial cutaneous nerve
**(D)** Medial antebrachial cutaneous nerve
**(E)** Radial nerve

**79.** A 54-year-old high school teacher has been diagnosed with coronary artery disease and is to undergo coronary bypass surgery. During the surgery, the thoracic surgeon decides to use the internal thoracic artery for one of the bypasses. Which of the following arteries gives rise to the internal thoracic artery?

**(A)** Axillary artery
**(B)** Superior epigastric artery
**(C)** Costocervical trunk
**(D)** Subclavian artery
**(E)** Ascending aorta

**80.** A 34-year-old singer has been diagnosed with thyroid cancer and consequently has a thyroidectomy. He has been hoarse ever since the surgery 8 weeks ago. It has been suspected that a nerve was injured during the operation. Which of the following nerves is most likely damaged?

**(A)** External laryngeal nerve
**(B)** Internal laryngeal nerve
**(C)** Recurrent laryngeal nerve
**(D)** Hypoglossal nerve
**(E)** Glossopharyngeal nerve

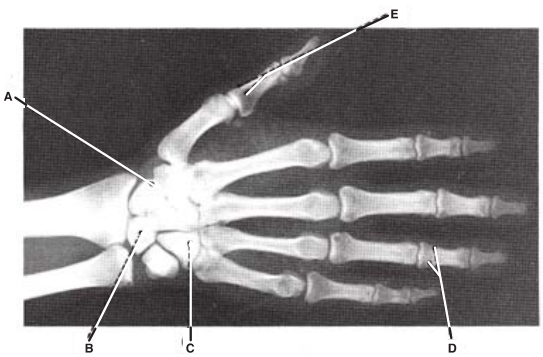
**81.** A 24-year-old woman presents with an asymptomatic fluid-filled cyst on the midline of the neck. She is diagnosed with a thyroglossal duct cyst. Which organ migrates through the thyroglossal duct during embryologic development?

**(A)** Inferior parathyroid glands
**(B)** Submandibular glands
**(C)** Thyroid gland
**(D)** Thymus
**(E)** Tongue

**82.** Otitis media is an infection in the middle ear cavity. From which embryologic structure does the middle ear cavity form?

**(A)** Pharyngeal groove 1
**(B)** Pharyngeal pouch 1
**(C)** Auricular hillocks
**(D)** Pharyngeal membrane 1
**(E)** Pharyngeal pouch 2

**Questions 83 to 87: Choose the appropriate lettered structure in the radiograph of the bones of the hand (see Figure below).**



**83.** Which bone articulates with the radius and triquetrum?

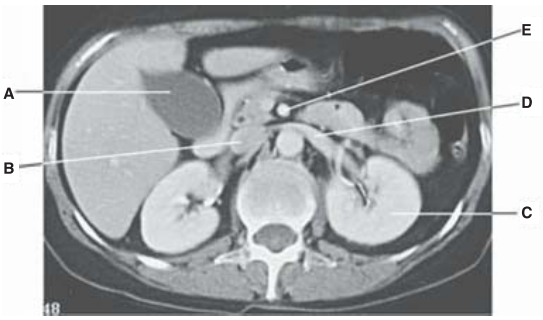
**84.** Fracture of which bone may cause paralysis of the flexor digiti minimi and opponens digiti minimi muscles?

**85.** Which is the site of attachment of the muscles that form the thenar eminence?

**86.** Which is the site of tendinous attachment of the flexor digitorum superficialis?

**87.** Fracture of which bone may cause a deep tenderness in the anatomic snuffbox?

**Questions 88 to 92: Choose the appropriate lettered structure in the CT scan of the abdomen (see Figure below).**




**88.** Thrombosis in which vessel causes a dilation of the left testicular vein?

**89.** Which structure concentrates and stores bile?

**90.** Laceration of which structure decreases blood flow in the middle colic artery?

**91.** Which structure produces and excretes urine?

**92.** Thrombosis in which vessel causes a dilation of the right suprarenal vein?

**Questions 93 to 97: Choose the appropriate lettered structure in the CT scan of the female pelvis (see Figure below).**




**93.** Which structure is a common site of uterine cancer?

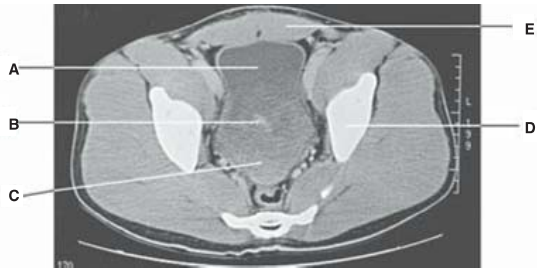
**94.** Which structure descends retroperitoneally on the psoas muscle and runs under the uterine artery?

**95.** Which structure has venous blood that returns to the portal and caval (systemic) venous systems?

**96.** Stimulation of parasympathetic nerve causes a contraction of the detrusor muscle in which structure?

**97.** When the lesser trochanter is fractured, which structure is paralyzed?

**Questions 98 to 100: Choose the appropriate lettered structure in the CT scan of the male pelvis (see Figure below).**

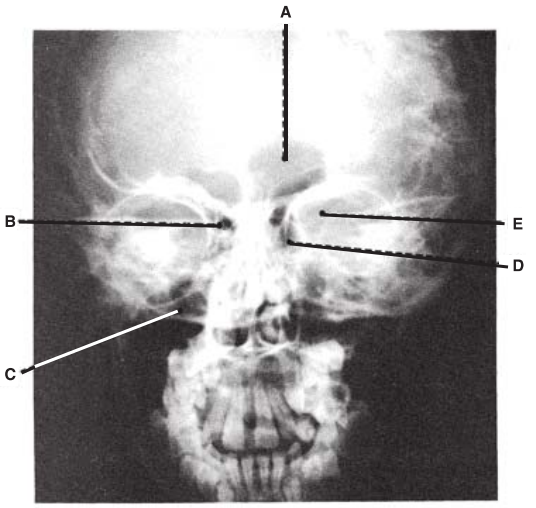



**98.** Which structure forms a medial boundary of the inguinal triangle?

**99.** Rupture of which structure impairs secretion of a fluid that produces the characteristic odor of semen?

**100.** Which structure receives the ejaculatory duct?

**Questions 101 and 102: Choose the appropriate lettered structure in the radiograph of the head (see Figure below).**




**101.** Which sinus opens into the hiatus semilunaris of the middle nasal meatus by way of the frontonasal duct or infundibulum?

**102.** Meningioma in which structure compresses the ophthalmic vein and trochlear nerve?

# Answers and Explanations

1. **The answer is B** (Chapter 7, Muscles of the Upper Limb: II.B). The axillary nerve runs posteriorly to the humerus, accompanying the posterior humeral circumflex artery through the quadrangular space and innervating the teres minor and deltoid muscles. None of the other nerves pass through the quadrangular space.
2. **The answer is D** (Chapter 7, Table 7.2 and Nerves of the Upper Limb: I.B.1 and F.4). Both the upper trunk and posterior cord of the brachial plexus are damaged. The abductors of the arm are the deltoid and supraspinatus muscles. The deltoid is innervated by the axillary nerve, which arises from the posterior cord of the brachial plexus. The supraspinatus is innervated by the suprascapular nerve, which arises from the upper trunk of the brachial plexus. The middle and lower trunks give rise to no branches. The lateral and medial cords supply no abductors of the arm.
3. **The answer is B** (Chapter 7, Table 7.5). The brachioradialis is innervated by the radial nerve and functions to flex the elbow. The flexor digitorum longus and extensor digitorum longus do not act in the elbow. The biceps brachii and brachialis muscles flex the elbow and are innervated by the musculocutaneous nerve.
4. **The answer is E** (Chapter 7, Nerves of the Upper Limb: I.F.5 and Clinical Correlates). The radial nerve innervates the extensor muscles of the hand; hence, Johnny could not extend his hand due to an injury to the radial nerve. Numbness would occur on the posterior aspects of the arm and forearm due to an injury to the radial nerve. The skin on the medial side of the forearm is innervated by the medial antebrachial cutaneous nerve; thus, numbness over the medial side of the forearm would not occur. The opponens pollicis, pronator teres, and pronator quadratus muscles are innervated by the median nerve. Therefore, inability to oppose the thumb or weakness in pronating the forearm would not occur. The abductors of the arm (deltoid and supraspinatus muscles) are innervated by the axillary nerve and upper trunk of the brachial plexus, respectively.
5. **The answer is E** (Chapter 7, Nerves of the Upper Limb: II.D and Clinical Correlates). Fracture of the medial epicondyle of the humerus might injure the ulnar nerve, which supplies the skin of the medial side of the hand; thus, a lesion of the ulnar nerve would cause no cutaneous sensation and lack of sweating in that area. The muscles involved in abduction of the hand are the flexor carpi radialis and the extensor carpi radialis longus and brevis, which are innervated by the median and radial nerves, respectively. Carpal tunnel syndrome and thenar atrophy result from a lesion of the median nerve, whereas wrist drop results from a lesion of the radial nerve.
6. **The answer is A** (Chapter 7, Table 7.6 and Nerves of the Upper Limb: II.D.2). The ulnar nerve innervates the dorsal interossei, which are the only abductors of the fingers. The little finger is abducted by the abductor digiti minimi, which is innervated by the ulnar nerve. The thumb is abducted by the abductor pollicis brevis and longus, which are innervated by the median and radial nerves, respectively. The musculocutaneous and axillary nerves do not supply the hand muscles.
7. **The answer is C** (Chapter 7, Nerves of the Upper Limb: II.B). The median nerve supplies the skin on the palmar aspect of the lateral three and one-half fingers and the dorsal side of the index finger, middle finger, and one-half of the ring finger. The median nerve innervates the pronator teres and pronator quadratus muscles and the thenar muscles. The axillary and musculocutaneous nerves do not supply the skin or muscles of the hand. The radial nerve does not innervate muscles of the hand but innervates the skin of the radial side of the hand and the radial two and one-half digits over the proximal phalanx. The ulnar nerve innervates not only the palmaris brevis, hypothenar muscles, adductor pollicis, dorsal and palmar interosseous, and medial two

lumbrical muscles but also the skin over the palmar and dorsal surfaces of the medial one-third of the hand and the skin of the little finger and the medial side of the ring finger.

8. **The answer is D** (Chapter 7, Table 7.2 and Nerves of the Upper Limb: I.F.2). The thoracodorsal nerve innervates the latissimus dorsi, which adducts, extends, and medially rotates the arm. The arm is abducted by the supraspinatus and laterally rotated by the infraspinatus, teres minor, and deltoid (posterior part) muscles. The scapula is elevated by the trapezium and levator scapulae muscles and adducted by the rhomboid and trapezius muscles.
9. **The answer is C** (Chapter 7, Nerves of the Upper Limb: III.B.2). The lateral antebrachial cutaneous nerve contains sympathetic postganglionic general visceral efferent (GVE) fibers, which have cell bodies located in the sympathetic chain ganglia, and general somatic afferent (GSA) fibers, which have cell bodies located in the dorsal root ganglia. Sympathetic nerve fibers are involved in sweating, whereas GSA fibers are involved in numbness and tingling in the skin.
10. **The answer is C** (Chapter 7, Table 7.6 and Nerves of the Upper Limb: II.D.2). This pianist is unable to pick up a music piece between the index and middle fingers because she cannot adduct her index finger and abduct the middle finger. The adductor of the index finger is the palmar interosseous muscle, which is innervated by the ulnar nerve. Abductors of the middle finger are dorsal interosseous muscles, which are innervated by the ulnar nerve. The other nerves do not innervate adductors or abductors of the fingers.
11. **The answer is D** (Chapter 7, Tables 7.5, 7.6, and Nerves of the Upper Limb: II.C.1 and D.2). The metacarpophalangeal joint of the ring finger is extended by the extensor digitorum, which is innervated by the radial nerve. This joint is flexed by the lumbrical and interossei muscles, abducted by the dorsal interosseous, and adducted by the palmar interosseous. The medial two lumbricals and both the dorsal and palmar interossei are innervated by the ulnar nerve. The median, musculocutaneous, and axillary nerves are not involved in movement of the metacarpophalangeal joint of the ring finger.
12. **The answer is A** (Chapter 7, Nerves of the Upper Limb: I.B.1). In Erb–Duchenne paralysis (or upper trunk injury), the nerve fibers in the roots of C5 and C6 of the brachial plexus are damaged. The infraspinatus, a lateral rotator muscle, is innervated by the suprascapular nerve (C5 and C6). All the other muscles, including the flexor carpi ulnaris, palmar interossei, adductor pollicis, and palmaris brevis muscles, are innervated by the ulnar nerve (C8 and T1).
13. **The answer is C** (Chapter 7, Nerves of the Upper Limb: II.B and Clinical Correlates). In carpal tunnel syndrome, structures entering the palm deep to the flexor retinaculum are compressed; these include the median nerve and the tendons of the flexor pollicis longus, flexor digitorum profundus, and flexor digitorum superficialis muscles. The flexor carpi radialis runs lateral to the carpal tunnel and inserts onto the bases of the second and third metacarpals. Structures entering the palm superficial to the flexor retinaculum include the ulnar nerve, ulnar artery, and palmaris longus tendon (which inserts onto the palmar aponeurosis).
14. **The answer is B** (Chapter 6, Joints and Ligaments of the Lower Limb: II.A.1.a). The anterior cruciate ligament of the knee joint prevents posterior displacement of the femur on the tibia and limits hyperextension of the knee joint. This ligament becomes taut when the knee is extended and lax when the knee is flexed. It inserts into the lateral femoral condyle posteriorly within the intercondylar notch.
15. **The answer is B** (Chapter 6, Table 6.5 and Nerves of the Lower Limb: I.F.1.b). The deep peroneal nerve innervates the dorsiflexors of the foot, which include the tibialis anterior, extensor hallucis longus, extensor digitorum longus, and peroneus tertius muscles. The plantar flexors include the triceps surae, tibialis posterior, flexor digitorum longus, and flexor hallucis longus, which are innervated by the tibial nerve, and peroneus longus and brevis, which are innervated by the superficial peroneal nerve. The toes are abducted and adducted by dorsal and plantar interosseous muscles, which are innervated by the medial and lateral plantar nerves. The foot is inverted by the tibialis anterior and posterior, triceps surae, and extensor hallucis longus, which are innervated by the tibial nerve.
16. **The answer is E** (Chapters 3, 4, and 6, Blood Vessels). An embolus from the deep veins of the lower limb would travel through the femoral vein, the iliac veins, the inferior vena cava, the right

atrium, the right ventricle, and the pulmonary trunk and into the pulmonary arteries, where it could obstruct and occlude these vessels. If not obstructed, blood from the pulmonary artery passes to the lungs, pulmonary veins, left atrium, left ventricle, ascending aorta, and coronary arteries; to the body tissues; and then to the venous system, including the renal veins and sinusoids of the liver.

17. **The answer is A** (Chapter 6, Muscles of the Lower Limb: VI.F.1 and Clinical Correlates). Flat foot is characterized by disappearance of the medial portion of the longitudinal arch, which appears completely flattened. The plantar calcaneonavicular (spring) ligament supports the head of the talus and the medial side of the longitudinal arch. The planar calcaneocuboid (short plantar) ligament supports the lateral portion of the longitudinal arch. The other ligaments support the ankle joint.

18. **The answer is D** (Chapter 6, Joints and Ligaments of the Lower Limb: II.A.1.b). The posterior cruciate ligament prevents forward displacement of the femur on the tibia when the knee is flexed. The anterior cruciate ligament prevents backward dislocation of the femur on the tibia when the knee is extended. The medial meniscus acts as a cushion, or shock absorber, and forms a more stable base for the articulation of the femoral condyle. The tibial and fibular collateral ligaments prevent medial and lateral displacement, respectively, of the two long bones.

19. **The answer is E** (Chapter 6, Muscles of the Lower Limb: VI.F.2). The lateral longitudinal arch is formed by the calcaneus, cuboid bone, and lateral two metatarsal bones, whereas the medial longitudinal arch of the foot is formed by the talus, calcaneus, navicular bone, cuneiform bones, and medial three metatarsal bones.

20. **The answer is E** (Chapter 5, Blood Vessels of the Pelvis: IX.B; Chapter 6, Blood Vessels of the Lower Limb: I.B–D). The median sacral artery terminates in the pelvis and is not involved in hip joint circulation. However, this joint receives blood from branches of the medial and lateral femoral circumflex, superior and inferior gluteal, and obturator arteries.

21. **The answer is D** (Chapter 6, Joints and Ligaments of the Lower Limb: II.A.1.d). The lateral meniscus, like the medial meniscus, lies outside the synovial cavity but within the joint cavity. However, the lateral meniscus is nearly circular, whereas the medial meniscus is C-shaped or forms a semicircle. The lateral meniscus is smaller than the medial meniscus and less frequently torn in injuries than the medial meniscus. In addition, the lateral meniscus is separated from the fibular collateral ligament by the tendon of the popliteal muscle, whereas the medial meniscus attaches to the tibial collateral ligament.

22. **The answer is C** (Chapter 1, Nervous System: IV.B). The ventral root of a thoracic spinal nerve contains sympathetic preganglionic fibers, which have cell bodies located in the lateral horn of the gray matter of the spinal cord, and general somatic efferent fibers, which have cell bodies located in the anterior horn of the gray matter of the spinal cord. The dorsal root ganglion contains cell bodies of GSA and general visceral afferent (GVA) fibers, and the sympathetic chain ganglion contains cell bodies of postganglionic sympathetic nerve fibers.

23. **The answer is B** (Chapter 3, Structures in the Posterior Mediastinum: II.B). The azygos vein receives the hemiazygos and accessory hemiazygos veins and drains into the superior vena cava. The internal thoracic vein drains into the subclavian vein. The right superior intercostal vein drains into the azygos vein, and the left superior intercostal vein drains into the left brachiocephalic vein.

24. **The answer is C** (Chapter 3, Pericardium and Heart: II.G.2). The great cardiac vein is accompanied by the anterior interventricular artery, which is a branch of the left coronary artery. The circumflex branch of the left coronary artery runs along with the coronary sinus. The right marginal artery is accompanied by the small cardiac vein, and the posterior interventricular artery is accompanied by the middle cardiac vein.

25. **The answer is B** (Chapter 3, Pericardium and Heart: II.F). The right and left coronary arteries arise from the ascending aorta, and reduced blood flow in the ascending aorta causes decreased blood flow in the coronary arteries. Blockage of the pulmonary trunk or coronary sinus minimally affects blood flow in the coronary arteries. Blockage of the origin of the aortic arch or the descending aorta increases blood flow in the coronary arteries.

26. **The answer is B** (Chapter 3, Structures in the Posterior Mediastinum: II.C.2). The right lymphatic duct drains the right sides of the thorax, upper limb, head, and neck, and the thoracic duct drains the rest of the body.

27. **The answer is D** (Chapter 3, Thoracic Wall: I.A.1). The third rib articulates with the body of the sternum rather than the manubrium. The manubrium of the sternum articulates with the body of the sternum, the first and second ribs, and the clavicle.

28. **The answer is D** (Chapter 1, Nervous System: V.A). Gray rami communicantes contain postganglionic sympathetic nerve fibers but contain no other nerve fibers.

29. **The answer is B** (Chapter 1, Nervous System: IV.B). The dorsal root contains both GSA and GVA fibers. The sympathetic trunk contains preganglionic sympathetic fibers. The greater splanchnic nerve and white rami communicantes contain GVA and preganglionic sympathetic fibers. The gray rami communicantes contain sympathetic postganglionic (GVE) fibers.

30. **The answer is C** (Chapter 3, Pericardium and Heart: II.G.1 and 6). Because the anterior cardiac vein drains directly into the right atrium, its diameter is unchanged. However, all other cardiac veins drain into the coronary sinus, and they are dilated.

31. **The answer is D** (Chapter 3, Mediastinum, Pleura, and Organs of Respiration: I). The mediastinum does not contain the lungs; it contains the thymus, esophagus, trachea, and heart.

32. **The answer is B** (Chapter 8, Table 8.9 and Oral Cavity and Palate: VII.C). The genioglossus protrudes the tongue. The hyoglossus and styloglossus muscles retract the tongue. The palatoglossus elevates the tongue. The geniohyoid elevates the hyoid bone and the floor of the mouth.

33. **The answer is D** (Chapter 3, Mediastinum, Pleura, and Organs of Respiration: VII). The pulmonary veins carry oxygenated blood, and the pulmonary artery carries deoxygenated blood. The right atrium (which contains the crista terminalis and pectinate muscle) and the right ventricle (which contains the septomarginal trabecula) carry deoxygenated blood.

34. **The answer is C** (Chapter 7, Table 7.6 and Nerves of the Upper Limb: II.D.2). The ulnar nerve arises from the lower trunk of the brachial plexus, which is formed by ventral primary rami of C8 and T1. Abduction of the index finger is done by the palmar interosseous muscle, which is innervated by the ulnar nerve. Lateral rotation of the arm is performed by the deltoid and teres minor muscles, which are innervated by the axillary nerve. The ring finger is extended by the extensor digitorum, which is innervated by the radial nerve. The forearm is flexed by the biceps brachii and brachialis muscles, which are innervated by the musculocutaneous nerve. Pronator teres and pronator quadratus muscles, which pronate the forearm, are innervated by the median nerve.

35. **The answer is C** (Chapter 5, Pelvis: VII.C and Clinical Correlates). The cardinal ligament provides the major support for the uterus. Therefore, the weakness of the cardinal ligament may result in uterine prolapse. The transversalis fascia, ovarian ligament, arcuate pubic ligament, and piriformis muscles do not support the uterus.

36. **The answer is A** (Chapter 4, Inguinal Region: A, B and C). An indirect inguinal hernia occurs lateral to the inferior epigastric vessels, whereas a direct inguinal hernia arises medial to these vessels. The other statements are not related to the indirect inguinal hernia.

37. **The answer is B** (Chapter 9, I.J). The vagus nerve supplies parasympathetic fibers to the thoracic and abdominal viscera, including the transverse colon. The descending colon, sigmoid colon, prostate, rectum, and other pelvic viscera are innervated by the pelvic splanchnic nerves.

38. **The answer is C** (Chapter 4, Gastrointestinal (GI) Viscera: V.C). The bile duct traverses the head of the pancreas; thus, a tumor located there could compress this structure. The duodenojejunal junction comes into contact with the tip of the uncinate process and the interior portion of the body of the pancreas. The other structures are not closely associated with the head of the pancreas.

39. **The answer is C** (Chapter 4, Gastrointestinal (GI) Viscera: VIII.A and B.1). The arterial supply to the pancreas is from both the celiac trunk and superior mesenteric artery. Other organs, including the liver, spleen, gallbladder, and stomach, receive blood from the celiac trunk.

40. **The answer is C** (Chapter 4, Gastrointestinal (GI) Viscera: IX.B). The superior rectal vein is part of the portal venous system. All the other veins belong to the systemic (caval) venous system.

41. **The answer is C** (Chapter 4, Gastrointestinal (GI) Viscera: IV.A Clinical Correlates). Hirschsprung disease (congenital megacolon) is abnormal dilatation of the lower colon proximal to the inactive segment and caused by the absence of enteric ganglia (parasympathetic postganglionic neuron cell bodies) in the myenteric plexus in the wall of the colon because of the failure of neural crest cells to migrate into the walls of the hindgut or colon. Abnormal rotation of the hindgut as they return to the abdominal cavity can cause volvulus or strangulation (twisting) of the gut. Abnormal formation of the urorectal septum or deficient (incomplete) separation of the cloaca results in anal agenesis.

42. **The answer is C** (Chapter 5, Perineal Region: II.B and V.A.3). The deep perineal space contains a segment of the dorsal nerve of the penis in males. The deep perineal space is bounded superiorly by the superior fascia and inferiorly by the inferior fascia (perineal membrane) of the urogenital diaphragm. The superficial transverse perineal muscles and the greater vestibular glands are found in the superficial perineal space. The Colles' fascia is the deep membranous layer of the superficial perineal fascia and forms the inferior boundary of the superficial perineal space.

43. **The answer is C** (Chapter 4, Gastrointestinal (GI) Viscera: III.C Clinical Correlates). The ileum is the most common site of Meckel diverticulum, which is located on the antimesenteric border of the ileum. This fingerlike pouch (diverticulum) is the remnant of the proximal portion of the embryonic yolk stalk (vitelline duct), which may be connected to the inner surface of the umbilicus via a fibrous cord or fistula. The other digestive organs are not associated with the development of the Meckel diverticulum.

44. **The answer is B** (Chapter 5, Pelvis: X.A.3). The inferior hypogastric (pelvic) plexus contains preganglionic parasympathetic fibers from the pelvic splanchnic nerves. The lumbar and sacral splanchnic nerves and the sacral sympathetic chain ganglia contain preganglionic sympathetic fibers. The vagus nerve does not supply parasympathetic nerve fibers to the pelvic organs.

45. **The answer is B** (Chapter 5, Pelvis: X.A.5). The urinary bladder receives parasympathetic fibers from the pelvic splanchnic nerve, not the vagus nerve. The greater, lesser, lumbar, and sacral splanchnic nerves contain sympathetic preganglionic fibers.

46. **The answer is C** (Chapter 5, Pelvis: IV.B). Carcinoma of the uterus can spread directly to the labium majus through the lymphatics that follow the round ligament of the uterus. This ligament extends from the uterus, enters the inguinal canal at the deep inguinal ring, emerges from the superficial inguinal ring, and merges with the subcutaneous tissue of the labium majus. The other ligaments do not reach the labium majus.

47. **The answer is C** (Chapter 5, Perineal Region: IV.B.1). The iliohypogastric nerve does not supply the scrotum. The scrotum is innervated by the anterior scrotal branch of the ilioinguinal nerve, the genital branch of the genitofemoral nerve, the posterior scrotal branch of the perineal branch of the pudendal nerve, and the perineal branch of the posterior femoral cutaneous nerve.

48. **The answer is C** (Chapter 5, Pelvis: I.C). Although the pubic crest forms a part of the pelvic inlet (pelvic brim), it does not contribute to the formation of the pubic outlet. The pelvic outlet (lower pelvic aperture) is bounded posteriorly by the sacrum and coccyx; laterally by the ischial tuberosities and sacrotuberous ligaments; and anteriorly by the pubic symphysis, the arcuate ligament, and the rami of the pubis and ischium.

49. **The answer is A** (Chapter 5, Pelvis: I.F). Compared with the male pelvis, the female pelvis is characterized by the presence of the oval inlet, larger outlet, larger pubic angle, shorter and wider sacrum, and wider and shallower cavity.

50. **The answer is E** (Chapter 2, Figure 2.2 and Spinal Cord and Associated Structures: III.B). The cauda equina is formed by dorsal and ventral roots of the lumbar and sacral spinal nerves. Thus, it is crushed at the level of the L5 vertebra, whereas the other structures are not. The vertebral artery, which arises from the subclavian artery, ascends through the transverse foramina of the upper six cervical vertebrae. The spinal cord ends at the level of the L2 vertebra. The filum terminale externus (filum of the dura) extends from the apex of the dura at the level of the S2 vertebra to the dorsum of the coccyx. The denticulate ligament, a lateral extension of the pia between the

dorsal and ventral roots of the spinal nerves, consists of 21 pairs of processes, the last one lying between the T12 and L1 spinal nerves.

51. **The answer is C** (Chapter 2, Table 2.2 and Structures of the Back: III.C). The suboccipital nerve supplies the suboccipital muscles, including the rectus capitis posterior major. The dorsal primary rami of the spinal nerves innervate the deep muscles of the back, including the splenius capitis and iliocostalis muscles. The spinal accessory nerve innervates the trapezius muscle, and the dorsal scapular nerve innervates the levator scapulae muscle; these are the superficial muscles of the back.

52. **The answer is C** (Chapter 2, Spinal Cord and Associated Structures: III Clinical Correlates). To obtain cerebrospinal fluid (CSF) contained in the subarachnoid space, the needle should penetrate the dura mater and arachnoid mater. The denticulate ligament, pia mater, annulus fibrosus, and nucleus pulposus should not be penetrated during a lumbar puncture.

53. **The answer is D** (Chapter 2, Table 2.1). The levator scapulae are attached to the transverse processes of the upper cervical vertebrae. All of the other muscles are attached to the spinous processes.

54. **The answer is B** (Chapter 8, Structures of the Neck: III.B.1.c). The transverse cervical nerve innervates the skin over the anterior cervical triangle; the great auricular nerve innervates the skin behind the auricle and over the parotid gland. The ansa cervicalis innervates the infrahyoid muscles, including the sternohyoid, sternothyroid, and omohyoid muscles. The superior laryngeal nerve divides into the internal laryngeal nerve, which supplies sensory fibers to the larynx above the vocal cord, and the external laryngeal nerve, which supplies the cricothyroid and inferior pharyngeal constrictor muscles.

55. **The answer is D** (Chapter 8, Structures of the Neck: IV.B.1.b and Clinical Correlates). The common carotid artery normally bifurcates into the external and internal carotid arteries at the level of the superior border of the thyroid cartilage. The thyroid isthmus crosses the second and third tracheal rings. The cricoid cartilage is at the level of CV6, and its lower border marks the end of the pharynx and larynx. The sternal angle is at the level of the intervertebral disk between TV4 and TV5, where the aortic arch begins and ends. The jugular notch is at the level of TV3.

56. **The answer is D** (Chapter 8, Table 8.4 and Temporal and Infratemporal Fossae: VI.A). The lateral pterygoid muscle is paralyzed because it inserts onto the articular disk and capsule of the temporomandibular joint. The temporalis muscle inserts onto the coronoid process, and the medial pterygoid and masseter muscles insert onto the medial and lateral surfaces of the ramus and angle of the mandible, respectively. The buccinator muscle inserts into the orbicularis oris at the angle of the mouth.

57. **The answer is C** (Chapter 9, I.C Clinical Correlates). Damage to the oculomotor nerve results in ptosis (drooping) of the eyelid because the levator palpebrae superioris is innervated by the oculomotor nerve. The trochlear nerve innervates the superior oblique, and the abducens nerve innervates the lateral rectus. The oculomotor nerve innervates the remaining ocular muscles. The facial nerve innervates the orbicularis oculi, which functions to close the eyelids.

58. **The answer is A** (Chapter 8, Skull and Cranial Cavity: VI.C.2.c). The superior petrosal sinus lies in the margin of the tentorium cerebelli. The falx cerebri contains the inferior and superior sagittal sinuses, and the falx cerebelli encloses the occipital sinus. The diaphragma sellae forms the dural roof of the sella turcica. The straight sinus runs along the line of attachment of the falx cerebri and tentorium cerebelli.

59. **The answer is A** (Chapter 8, Skull and Cranial Cavity: VI.B.2). Arachnoid granulations are tuftlike collections of highly folded arachnoid that project into the superior sagittal sinus and the lateral lacunae, which are lateral extensions of the superior sagittal sinus. They absorb the CSF into dural sinuses and often produce erosion or pitting of the inner surface of the calvaria. The CSF is produced in the ventricles of the brain.

60. **The answer is E** (Chapter 8, Skull and Cranial Cavity: VII.B). The great cerebral vein of Galen and the inferior sagittal sinus unite to form the straight sinus.

61. **The answer is B** (Chapter 8, Table 8.8). The tendon of the tensor veli palatini curves around the pterygoid hamulus to insert onto the soft palate. The tensor tympani inserts onto the handle of the malleus, the levator veli palatini inserts onto the soft palate, the superior pharyngeal

constrictor inserts onto the median raphe and the pharyngeal tubercle, and the stylohyoid inserts onto the body of the hyoid.

62. **The answer is C** (Chapter 8, Temporal and Infratemporal Fossae: VI.B). The pterygomandibular raphe serves as a common origin for the superior pharyngeal constrictor and buccinator muscles. None of the other choices are involved with the pterygomandibular raphe.

63. **The answer is D** (Chapter 9, I.E). The tensor veli palatini is innervated by the trigeminal nerve, the levator veli palatini and palatoglossus are innervated by the vagus nerve, the cricothyroid is innervated by the external laryngeal branch of the superior laryngeal nerve, and the geniohyoid muscle is innervated by the first cervical nerve through the hypoglossal nerve.

64. **The answer is E** (Chapter 8, Oral Cavity and Palate: V.A). The mylohyoid muscle indents the submandibular gland and divides it into superficial and deep parts. The deep portion of the gland is located between the mylohyoid muscle laterally and the hyoglossus and styloglossus muscles medially. Posteriorly, the gland lies against the posterior digastric and stylohyoid muscles.

65. **The answer is B** (Chapter 8, Pharynx and Tonsils: II.A). The auditory (Eustachian) tube connects the nasopharynx with the middle ear cavity. The vestibule and semicircular canals are parts of the inner ear. The external ear and inner ear communicate with the nasopharynx.

66. **The answer is D** (Chapter 8, Table 8.4). The action of the lateral pterygoid muscles opens the jaws. The medial pterygoid, masseter, and temporalis muscles are involved in closing the jaws. The buccinator presses the cheek to keep it taut.

67. **The answer is C** (Chapter 9, I.I.1). Parasympathetic preganglionic fibers in the lesser petrosal nerve enter the otic ganglion where they synapse, and postganglionic parasympathetic fibers join the auriculotemporal nerve to supply the parotid gland for secretion of saliva. The other conditions are not caused by a lesion of the lesser petrosal nerve.

68. **The answer is D** (Chapter 9, I.J). The musculus uvulae are innervated by the vagus nerve. A lesion of the vagus nerve results in deviation of the uvula toward the opposite side of the injury. An injury of the right vagus nerve causes paralysis of the right uvular muscle, which means that the uvula deviates toward the left. A lesion of the hypoglossal nerve causes deviation of the tongue toward the injured side on protrusion. A lesion of the trigeminal nerve (mandibular division) causes paralysis of the mastication muscles.

69. **The answer is D** (Chapter 8, Pharynx and Tonsils: VI.B and Clinical Correlates). The superior thyroid artery does not supply the palatine tonsil. The palatine tonsil receives blood from the lesser palatine branch of the maxillary artery, the ascending palatine branch of the facial artery, the dorsal lingual branches of the lingual artery, and the ascending pharyngeal artery.

70. **The answer is D** (Chapter 9, I.G and 1). The parasympathetic secretomotor fibers for mucous glands in the nasal cavity run in the facial nerve, the greater petrosal nerve, the nerve of the pterygoid canal, and the pterygopalatine ganglion. The lesser petrosal nerve contains parasympathetic preganglionic fibers for the parotid gland. The facial nerve in the facial canal contains parasympathetic fibers for submandibular and sublingual salivary glands but not for nasal mucosal glands. The superior cervical ganglion and the deep petrosal nerve contain sympathetic postganglionic nerve cell bodies and/or fibers, which supply blood vessels in the lacrimal gland and the nasal and palate mucosa.

71. **The answer is A** (Chapter 2, Spinal Cord and Associated Structures: I Clinical Correlates). Tethered cord syndrome is frequently associated with spina bifida occulta. Spina bifida occulta is a condition caused by failure of the vertebral arch to fuse, with no protrusion of the spinal cord and the meninges. Meningomyelocele is a protrusion of the spinal cord and the meninges through the unfused arch of the vertebra. Kyphosis is an abnormal accentuation of lumbar curvature. Herniated disk represents a protrusion of the nucleus pulposus through the annulus fibrosus of the intervertebral disk into the intervertebral foramen or into the vertebral canal, compressing the spinal nerve roots. Scoliosis is a lateral deviation of the spine resulting from unequal growth of the vertebral column.

72. **The answer is D** (Chapter 6, Nerves of the Lower Limb: I.B and F). The quadratus femoris muscles—the flexors of the thigh—are innervated by the femoral nerve, which originates from the spinal cord at L2 to L4. In contrast, the hamstring muscles—the extensors of the thigh—are

innervated by the sciatic nerve, which originates from L4 to S3. Therefore, the lesion occurs at the level of L5 (between L4 and S3).

73. **The answer is B** (Chapter 7, Nerves of the Upper Limb: II.B). The median nerve enters the palm of the hand through the carpal tunnel deep to the flexor retinaculum, giving off a muscular branch (recurrent branch) to the thenar muscles, including the abductor pollicis brevis, flexor pollicis brevis, and opponens pollicis. The ulnar nerve enters the hand superficial to the flexor retinaculum and lateral to the pisiform bone, supplying the hypothenar muscles and adductor pollicis. The patient can adduct his thumb but not oppose it. Therefore, the median nerve is injured. The radial nerve and the anterior and posterior interosseous nerves do not supply the muscle that opposes the thumb.

74. **The answer is B** (Chapter 6, Nerves of the Lower Limb: I.F.2 and Clinical Correlates). The tibial nerve innervates the triceps surae, plantaris, and posterior tibialis, which plantar flex; innervates the flexor digitorum longus and brevis, flexor hallucis longus and brevis, and flexor digiti minimi brevis, which flex toes; and supplies sensory innervation on the sole of the foot. The common peroneal nerve divides into the deep peroneal nerve, which innervates muscles of the anterior compartment that dorsiflex the foot, and the superficial peroneal nerve, which innervates the peroneus longus and brevis that dorsiflex the foot. The saphenous nerve is a cutaneous nerve and does not supply the muscles.

75. **The answer is C** (Chapter 8, Orbit: VI Clinical Correlates). Lung cancer in the apex of the lung compresses the cervical sympathetic trunk and stellate ganglion, causing ptosis, miosis, enophthalmos, and anhidrosis, which are symptoms of Horner syndrome. Injury of the other nerves does not cause ptosis, miosis, enophthalmos, or anhidrosis.

76. **The answer is D** (Chapter 7, Nerves of the Upper Limb: II.D and Clinical Correlates). The ulnar nerve supplies sensation to the fourth and fifth digits and innervates the hypothenar muscles, the dorsal interosseous muscles that abduct the fingers, and the medial half of the flexor digitorum profundus (to the ring and little fingers or fourth and fifth digits) that flexes the distal interphalangeal joints. Claw hand is a condition in which the ring and little fingers are hyperextended at the metacarpophalangeal joints and flexed at the interphalangeal joints. The axillary and anterior interosseous nerves do not supply the hand. Injury to the radial nerve results in wrist drop. Injury to the median nerve causes an ape hand (flattening of the thenar eminence).

77. **The answer is D** (Chapter 7, Nerves of the Upper Limb: I.A.2 Clinical Correlates). The long thoracic nerve innervates the serratus anterior muscle. Paralysis of this muscle causes a "winged scapula," in which the vertebral or medial border and inferior angle of the scapula protrude away from the thorax. Other nerves do not supply the serratus anterior muscle.

78. **The answer is C** (Chapter 7, Nerves of the Upper Limb: I.D.2). The medial brachial cutaneous nerve supplies the skin on the medial aspect of the arm. The axillary nerve supplies the skin of the lateral side of the arm. The musculocutaneous nerve supplies the lateral side of the forearm as the lateral antebrachial cutaneous nerve. The medial antebrachial cutaneous nerve supplies the medial aspect of the forearm. The radial nerve gives off the posterior brachial and posterior antebrachial cutaneous nerves.

79. **The answer is D** (Chapter 3, Thoracic Wall: V.B). The internal thoracic or internal mammary artery arises from the subclavian artery. Other arteries do not give rise to the internal thoracic artery. The superior epigastric artery is a branch of the internal thoracic artery.

80. **The answer is C** (Chapter 9, I.J.8). The recurrent laryngeal nerve runs behind the thyroid gland in a groove between the trachea and esophagus and is vulnerable to injury during thyroidectomy. This nerve innervates all of the laryngeal muscles, except the cricothyroid muscle, which is innervated by the external laryngeal nerve. The internal laryngeal nerve is sensory to the larynx above the vocal cord. The hypoglossal and glossopharyngeal nerves do not supply the larynx and are not closely associated with the thyroid gland.

81. **The answer is C** (Chapter 8, Deep Neck and Prevertebral Region: IV.A and Clinical Correlates). Thyroglossal duct cyst results from the incomplete obliteration of the duct through which the thyroid gland migrates to the root of the neck. Inferior parathyroid glands and the thymus migrate in the neck lateral to the thyroglossal duct. The tongue and submandibular glands form and remain in the floor of the primitive mouth.

82. **The answer is B** (Chapter 8, Pharynx and Tonsils: VIII.B.1). The middle ear cavity develops from pharyngeal pouch 1. Pharyngeal groove 1 forms the external auditory canal; auricular hillocks form the auricle; pharyngeal membrane 1 forms the tympanic membrane; and pharyngeal pouch 2 forms the crypts of the palatine tonsil.
83. **The answer is B** (Chapter 7, Figure 7.2). The lunate bone articulates with the radius and triquetrum.
84. **The answer is C** (Chapter 7, Table 7.6). The hook of the hamate provides attachment for the flexor digiti minimi brevis and opponens digiti minimi muscles. Therefore, its fracture may cause paralysis of these muscles.
85. **The answer is E** (Chapter 7, Table 7.6). The base of the proximal phalanx of the thumb is the site of attachment for the flexor pollicis brevis, which, along with the opponens pollicis, forms the thenar eminence. It is also the site of attachment for the adductor pollicis brevis.
86. **The answer is D** (Chapter 7, Table 7.4). The middle phalanx of the ring finger is the site of attachment for the flexor digitorum superficialis.
87. **The answer is A** (Chapter 7, Figure 7.2 and Muscles of the Upper Limb: IV.J). The scaphoid bone forms the floor of the anatomic snuffbox, and its fracture may cause a deep tenderness. When fractured, the proximal fragment may undergo avascular necrosis because the blood supply is interrupted.
88. **The answer is D** (Chapter 5, Pelvis: VI.A). The left renal vein receives the left testicular vein.
89. **The answer is A** (Chapter 4, Gastrointestinal (GI) Viscera: V.B). The gallbladder receives bile and concentrates and stores it.
90. **The answer is E** (Chapter 4, Gastrointestinal (GI) Viscera: VIII.B). The superior mesenteric artery gives off the middle colic artery.
91. **The answer is C** (Chapter 4, Retroperitoneal Viscera, Diaphragm, and Posterior Abdominal Wall: I.A). The kidney produces and excretes urine.
92. **The answer is B** (Chapter 4, Retroperitoneal Viscera, Diaphragm, and Posterior Abdominal Wall: I.C). The right suprarenal vein drains into the inferior vena cava. However, the left renal vein receives the left suprarenal vein.
93. **The answer is D** (Chapter 5, Pelvis: VII.C.4 and Clinical Correlates). The uterine cervix is the common site of uterine cancer.
94. **The answer is B** (Chapter 5, Pelvis: V.A). The ureter descends retroperitoneally on the psoas muscle in the abdomen and runs under the uterine artery in the pelvis.
95. **The answer is C** (Chapter 5, Pelvis: VIII.A). The rectum returns its venous blood to the portal vein via the superior rectal vein and to the inferior vena cava (caval or systemic venous system) via the middle and inferior rectal veins.
96. **The answer is A** (Chapter 5, Pelvis: V.B). The detrusor muscle in the wall of the bladder is innervated by sympathetic nerves.
97. **The answer is E** (Chapter 6, Table 6.3). The iliacus muscle, together with the psoas major muscle, inserts onto the lesser trochanter.
98. **The answer is E** (Chapter 4, Anterior Abdominal Wall: IV.A). The margin of the rectus abdominis forms the medial boundary of the inguinal triangle.
99. **The answer is C** (Chapter 5, Pelvis: VI.F). The prostate gland secretes a fluid that produces the characteristic odor of semen.
100. **The answer is B** (Chapter 5, Pelvis: VI.F). The seminal colliculus or verumontanum of the prostatic urethra receives the ejaculatory duct.
101. **The answer is A** (Chapter 8, Nasal Cavity and Paranasal Sinuses: I.D.3). The frontal sinus opens into the hiatus semilunaris of the middle nasal meatus by way of the frontonasal duct or infundibulum.
102. **The answer is E** (Chapter 8, Orbit: I.C.1). The superior orbital fissure transmits the ophthalmic vein, trochlear nerve, and other structures.